Bharath Vedantha Desikan

M.S. Robotics & Autonomous Systems, Arizona State University

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Summary

Robotics researcher focused on perception and planning for aerial and aquatic robots. Experience across ROS 2 system development, visual-inertial odometry, SLAM and LiDAR mapping, teleoperation, and uncertainty-aware planning, with peer-reviewed robotics publications.

Experience

Research Aide — Robotics Perception & Simulation, NASA STTR Jun – Jul 2026
DREAMS Lab, Arizona State University *Tempe, AZ*

- Built high-fidelity ROS 2 simulation frameworks with a realistic "Reef World" coral-reef environment to test and validate underwater drone (AUV) operations.
- Developed perception pipelines using visual-inertial odometry (VIO) for GPS-denied localization; modeled and analyzed positional drift and sensor-fusion challenges under realistic underwater conditions.
- Evaluated navigation robustness for autonomous underwater vehicles under noisy state estimates and limited on-board compute.

Research Intern Jan – Jun 2024
Shimono Laboratory, Yokohama National University *Yokohama, Japan*

- Designed and implemented a ROS 2 framework for bilateral teleoperation of Universal Robots (UR) manipulators, enabling real-time force-feedback control between leader and follower arms.
- Built the full ROS 2 stack — nodes, controllers, and communication — for low-latency exchange of position and force commands with haptic feedback.
- Tuned control loops for stability and transparency and validated performance under communication delay.

ROS 2 Intern May – Jun 2023
ROBIS, Motherson Group *India*

- Facilitated motion planning and control system design for an AGV switchover from PLC-based automation to ROS 2.
- Integrated LiDARs with ROS 2 to initiate SLAM-based mapping, generating occupancy maps for autonomous AGV navigation.

Publications & Thesis

Field Estimation and Adaptive Sampling under Positional Uncertainty
B. Vedantha Desikan. M.S. Thesis, Advisor: Prof. J. Das — *Arizona State University, 2026* *thesis link* M.S. Thesis

Depth-Variance Guided Next-Best-View Planning for Active 3D Gaussian Splatting Reconstruction
B. Vedantha Desikan, R. Staggers Jr., K. Prashar, C. Torre, J. Das — *IEEE CASE 2026* *paper link* Presented

Low-Cost ASV for High-Resolution Spatio-Temporal Aquatic Field Reconstruction via Dynamic Kernels
B. Vedantha Desikan[*], R. Staggers Jr.[*], J. Das — *IEEE ICRA 2025 Workshop on Robots in the Wild* Published

[*]equal contribution

Active Gaussian Splatting for Autonomous Reconstruction Using D-Optimal Next Best View Planning
K. Prashar, C. Torre, R. Staggers Jr., **B. Vedantha Desikan**, J. Das — *IEEE CASE 2026* *paper link* Presented

Education

M.S. Robotics & Autonomous Systems 2024 – 2026
Arizona State University *GPA: 3.74/4.00*

B.Tech. Mechatronics & Automation 2020 – 2024
Vellore Institute of Technology, Chennai

Skills

Robotics: ROS 2, PX4 Autopilot, Gazebo, SLAM, LiDAR Mapping, Informative Path Planning, Next-Best-View Planning

ML & Perception: Visual-Inertial Odometry, Gaussian Processes, 3D Gaussian Splatting, Uncertainty-Aware Planning, PyTorch, GPyTorch, OpenCV

Systems: Low-SWaP Hardware, Bandwidth-Constrained Autonomy, GPS-Denied Operation

Tools: Python, CUDA, MATLAB, Linux (Ubuntu), Docker, Git